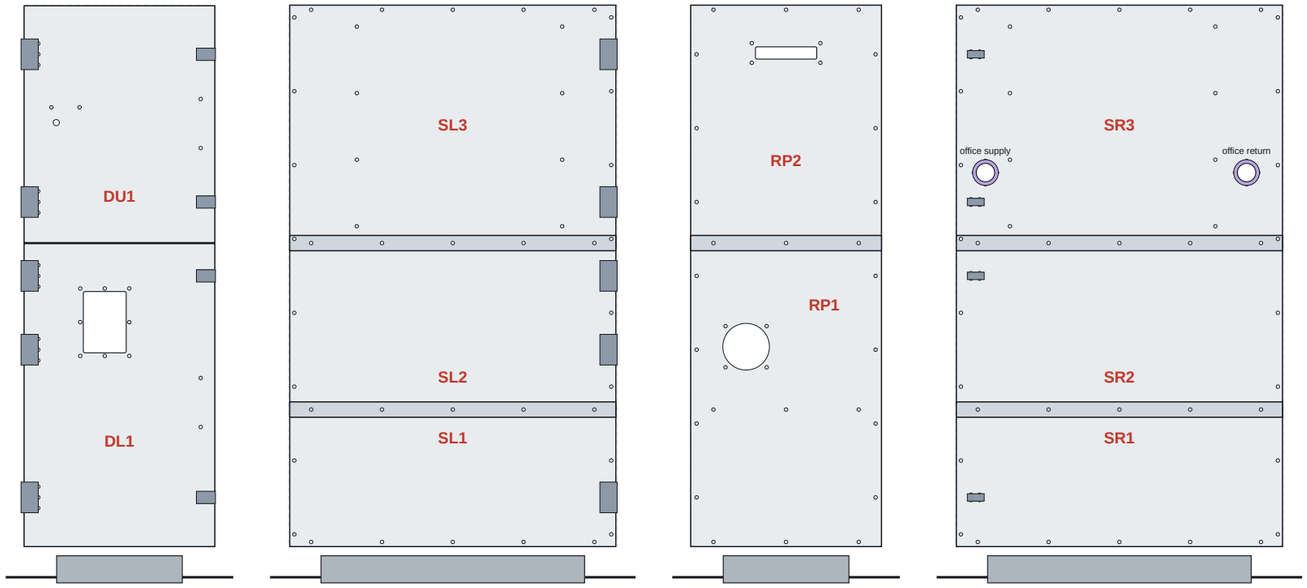
SKIN LAYOUT - EXTERIOR ELEVATIONS 1:20 (pieces as seen from outside)



FRONT | LEFT SIDE | REAR | RIGHT SIDE

TOP (front edge at bottom)

Grey bars: hinges (front + left), latches + keepers (front + right).
Purple: office air ports on SR3 - P9 adaptor's D60 duct spigot (rev C).
Strips JS/JR cover the skin joints on the mid and shelf rails.
Skirts SK1-SK3 sit under the base rails between the castor pads.
Dots: D8 screw holes on the frame rivnuts (weld pack sheet 6).
Every frame rivnut is matched by a hole (checked by sheetmetal.py).

PARTS (one DXF each)

ID	QTY	PART	t	FLAT	HOLES	BENDS
SL1	1	Side skin left - bottom - HOLD M3	1.2	1064 x 446	12	0
SL2	1	Side skin left - middle - HOLD M3	1.2	1064 x 542	10	0
SL3	1	Side skin left - top	1.2	1064 x 776	27	0
SR1	1	Side skin right - bottom - HOLD M3	1.2	1064 x 446	13	0
SR2	1	Side skin right - middle - HOLD M3	1.2	1064 x 542	8	0
SR3	1	Side skin right - top	1.2	1064 x 776	39	0
JS1	2	Joint strip, side skins - mid rail	1.2	1064 x 50	5	0
JS2	2	Joint strip, side skins - shelf rail	1.2	1064 x 50	7	0
RP1	1	Rear skin - bottom - HOLD M3	1.2	622 x 990	19	0
RP2	1	Rear skin - top	1.2	622 x 776	13	0
JR1	1	Joint strip, rear skin	1.2	622 x 50	3	0
TP1	1	Top skin	1.2	622 x 1072	13	0
DL1	1	Door, lower	1.2	622 x 988	23	0
DU1	1	Door, upper	1.2	622 x 773	15	0
DA1	1	Stiffener lower door, left vertical	1.2	941 x 38	9	1
DA2	1	Stiffener lower door, right vertical	1.2	941 x 38	6	1
DA3	1	Stiffener upper door, left vertical	1.2	727 x 38	6	1
DA4	1	Stiffener upper door, right vertical	1.2	727 x 38	6	1
DA5	4	Stiffener horizontal	1.2	510 x 38	0	1
KB1	4	Latch keeper bracket	1.2	83 x 40	2	1
WR1	1	Window retainer frame	1.2	180 x 240	8	0
SK1	1	Plinth skirt, front	1.2	410 x 116	3	1
SK2	1	Plinth skirt, rear	1.2	410 x 116	3	1
SK3	2	Plinth skirt, side	1.2	860 x 116	4	1
DT1	1	Drip tray	1.2	603 x 1053	1	4
PD1	4	Castor pad	8	100 x 100	4	0
PC1	4	Post top cap	3	30 x 30	0	0
RR1	4	19in rail, 16U	2	721 x 46	51	1

46 pieces, 78 kg. Largest blank 1064 x 776 (limit 1200 x 800).

Blanks 1200x800: steel_1.2mm x11, SS304_1.2mm x1, plate_8mm x1, plate_3mm x1, steel_2mm x1

ORDERING

Send cad/dxf/cut/ - each file is named ID_thickness_material_qty,
e.g. SL3_1.2mm_steel_x1.dxf - plus bom/sheet_metal_parts.csv.
Or cut the nested blanks in cad/dxf/nest/ on a 1200x800 bed.
Folding: 11 parts, 24 bends (flat blanks: R = t, K 0.33).
Tap the four PD1 pads M8 after cutting. Coat after folding.
HOLD RP1 SL1 SL2 SR1 SR2 until tape check M3: they move with the mid rail.

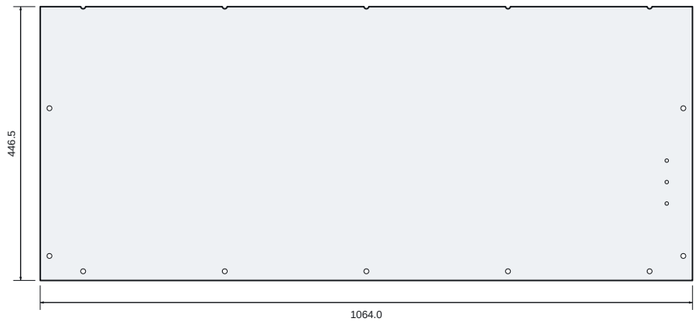
NOTES

1 DXF (cad/dxf/cut) is the master geometry, 1:1 mm. info/ adds bend lines, ID etch and notes on layers.
2 Skins 1.2 mm steel (Zincanneal/CR4), powder coat both sides, then bond 3 mm MLV inside (D8 at the frame screws: it is the gasket).
3 Frame screws: M6 x 20 flanged button head through D8 (+-1 mm frame tolerance) into the frame
rivnuts. Rivets: 4.8 mm blind, D5 holes. Deburr all holes; mask nothing - coat before rivnuts.
4 Folded parts: flat blanks for R = t inside, K 0.33. Check one test bend on your brake first.
5 Split skins meet on the rail centreline with a 1 mm gap and R4 notches round each strip screw; the strip
clamps both edges (sealant under). Detail J, sheet 5.

PROJECT	SRA-16 Silent AC Rack - laser-cut sheet-metal parts
TITLE	Skin layout, parts list
DOC NO.	CP-SRA16-SMP-001 SHEET 1 OF 6 REV C
SCALE	1:20 (elevations) @ A3 UNITS mm LAYOUT v1.3.1
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	FOR CUTTING - DXF is the master; confirm hardware (sheet 5)

SL1 x1 — Side skin left - bottom — 1:10

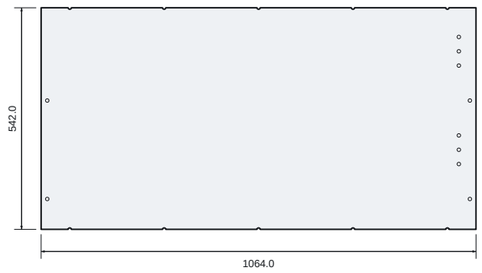
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



9 x D8 M6 screw to frame rivnut
3 x D5 rivet, hinge frame leaf
5 x R4 edge notch, joint strip screw

SL2 x1 — Side skin left - middle — 1:15

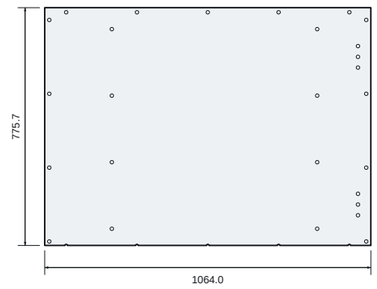
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



4 x D8 M6 screw to frame rivnut
6 x D5 rivet, hinge frame leaf
10 x R4 edge notch, joint strip screw

SL3 x1 — Side skin left - top — 1:20

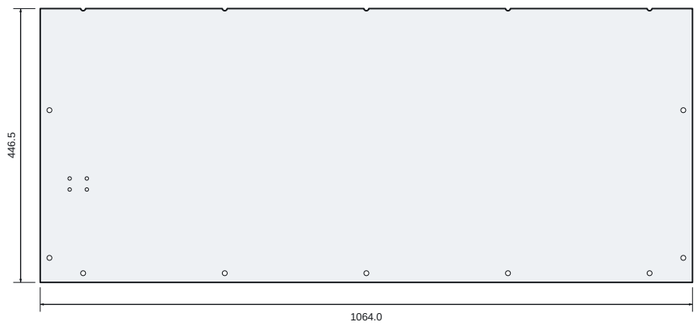
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



21 x D8 M6 screw to frame rivnut
6 x D5 rivet, hinge frame leaf
5 x R4 edge notch, joint strip screw

SR1 x1 — Side skin right - bottom — 1:10

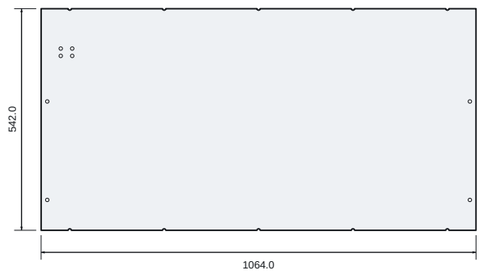
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



9 x D8 M6 screw to frame rivnut
4 x D5 rivet, toggle latch base
5 x R4 edge notch, joint strip screw

SR2 x1 — Side skin right - middle — 1:15

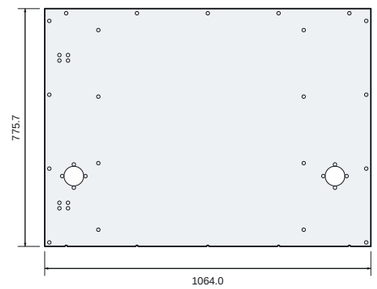
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



4 x D8 M6 screw to frame rivnut
4 x D5 rivet, toggle latch base
10 x R4 edge notch, joint strip screw

SR3 x1 — Side skin right - top — 1:20

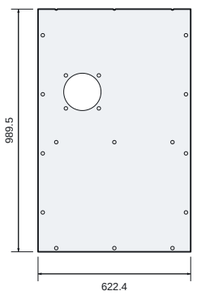
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



21 x D8 M6 screw to frame rivnut
8 x D5 rivet, toggle latch base
1 x D64 office air supply port D64
4 x D3.4 M3, office supply port

RP1 x1 — Rear skin - bottom — 1:25

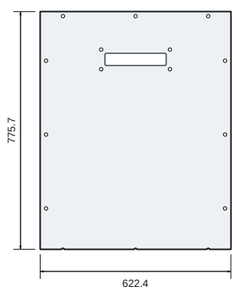
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



14 x D8 M6 screw to frame rivnut
1 x D152 exhaust duct D150
4 x D5.5 M5, exhaust spigot flange
3 x R4 edge notch, joint strip screw

RP2 x1 — Rear skin - top — 1:20

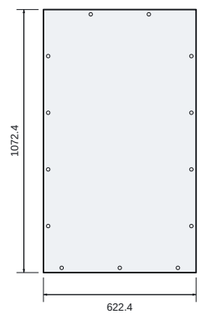
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



9 x D8 M6 screw to frame rivnut
4 x D4.5 M4, brush cable grommet
cable entry 200x40
3 x R4 edge notch, joint strip screw

TP1 x1 — Top skin — 1:25

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides

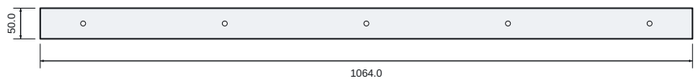


13 x D8 M6 screw to frame rivnut

PROJECT	SRA-16 Silent AC Rack - laser-cut sheet-metal parts
TITLE	Flat patterns - side, rear and top skins
DOC NO.	CP-SRA16-SMP-001 SHEET 2 OF 6 REV C
SCALE	as noted per part @ A3 UNITS mm LAYOUT v1.3.1
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	FOR CUTTING - DXF is the master; confirm hardware (sheet 5)

JS1 x2 — Joint strip, side skins - mid rail — 1:10

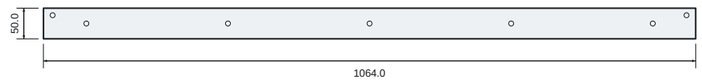
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



5 x D8 M6 screw to frame rivnut

JS2 x2 — Joint strip, side skins - shelf rail — 1:10

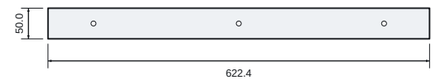
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



7 x D8 M6 screw to frame rivnut

JR1 x1 — Joint strip, rear skin — 1:10

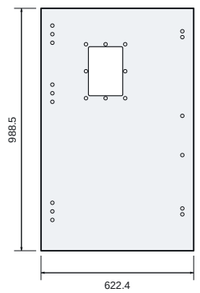
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



3 x D8 M6 screw to frame rivnut

DL1 x1 — Door, lower (AC bay) — 1:25

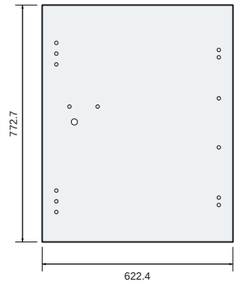
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



9 x D5 rivet, hinge door leaf
4 x D5 rivet, latch keeper KB1
2 x D5.5 M5, pull handle
8 x D4.5 M4, window clamp

DU1 x1 — Door, upper (rack) — 1:20

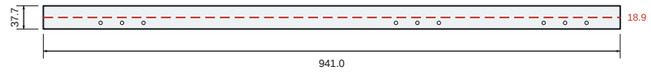
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035 both sides



6 x D5 rivet, hinge door leaf
4 x D5 rivet, latch keeper KB1
2 x D5.5 M5, pull handle
2 x D6 M4 rivnut, touchscreen pod

DA1 x1 — Door stiffener, lower door, left vertical — 1:10

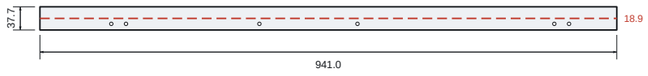
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



9 x D5 rivet, hinge

DA2 x1 — Door stiffener, lower door, right vertical — 1:10

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



4 x D5 rivet, latch keeper
2 x D5.5 M5, pull handle screw

DA3 x1 — Door stiffener, upper door, left vertical — 1:10

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



6 x D5 rivet, hinge

DA4 x1 — Door stiffener, upper door, right vertical — 1:10

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035

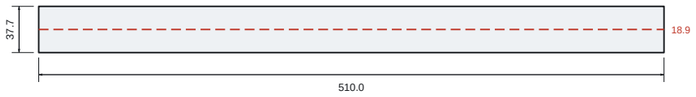


4 x D5 rivet, latch keeper
2 x D5.5 M5, pull handle screw

PROJECT	SRA-16 Silent AC Rack - laser-cut sheet-metal parts
TITLE	Flat patterns - doors, joint strips, door stiffeners
DOC NO.	CP-SRA16-SMP-001 SHEET 3 OF 6 REV C
SCALE	as noted per part @ A3 UNITS mm LAYOUT v1.3.1
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	FOR CUTTING - DXF is the master; confirm hardware (sheet 5)

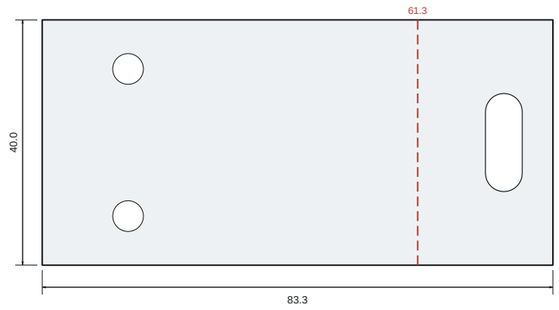
DA5 x4 — Door stiffener, horizontal (top/bottom, both doors) — 1:5

Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



KB1 x4 — Latch keeper bracket — 1:1

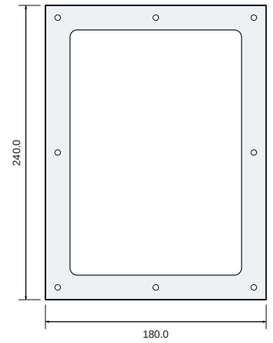
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



2 x D5 rivet to door + stiffener
latch catch slot

WR1 x1 — Window retainer frame — 1:5

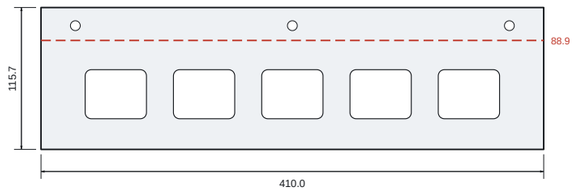
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



8 x D4.5 M4, window clamp
window opening

SK1 x1 — Plinth skirt, front — 1:5

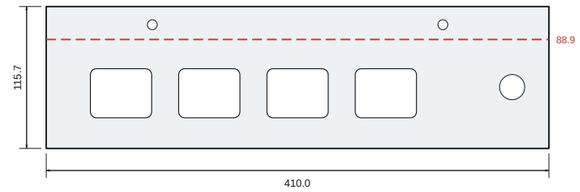
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



3 x D8 M6 screw up into base-rail rivnut
air slot (x5)

SK2 x1 — Plinth skirt, rear (drain outlet) — 1:5

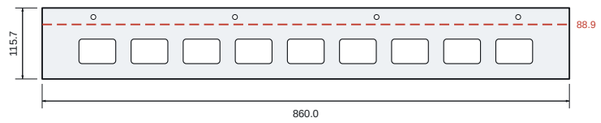
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



2 x D8 M6 screw up into base-rail rivnut
1 x D20.5 drain barb bulkhead
air slot (x4)

SK3 x2 — Plinth skirt, side (right: same part turned round) — 1:10

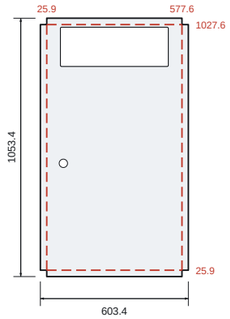
Steel sheet (Zincanneal or CR4) 1.2 - powder coat RAL 7035



4 x D8 M6 screw up into base-rail rivnut
air slot (x9)

DT1 x1 — Drip tray (304 stainless pan) — 1:25

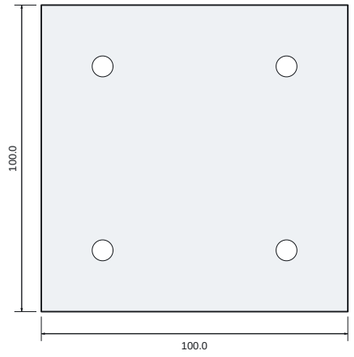
Stainless 304 sheet, 2B 1.2 - as cut, TIG-weld corners, passivate



1 x D34 25 mm tank bulkhead
room-air inlet (riser)

PD1 x4 — Castor pad (weld pack mark PL1) — 1:2

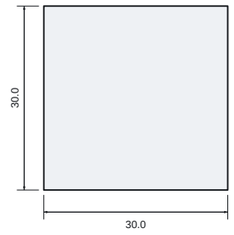
Steel plate (grade 250) 8.0 - welded to frame, coated with it



4 x D6.8 tap M8 through

PC1 x4 — Post top cap (weld pack mark PL2) — 1:1

Steel plate (grade 250) 3.0 - welded to frame, coated with it



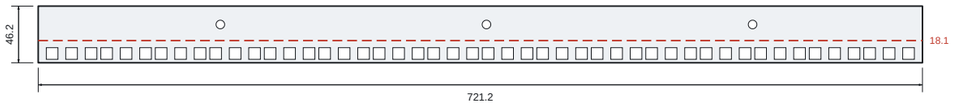
PROJECT	SRA-16 Silent AC Rack - laser-cut sheet-metal parts
TITLE	Flat patterns - stiffeners, keepers, window, skirts, tray, plates
DOC NO.	CP-SRA16-SMP-001 SHEET 4 OF 6 REV C
SCALE	as noted per part @ A3 UNITS mm LAYOUT v1.3.1
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	FOR CUTTING - DXF is the master; confirm hardware (sheet 5)

RR1 x4

19in rail, 16U (EIA-310 square holes)

1:5

Steel sheet (CR4) 2.0 - zinc plate (keep cage-nut holes clear)



3 x D7 M6 bolt through rail spacer into upright rivnut
48 x 9.5 sq cage nut

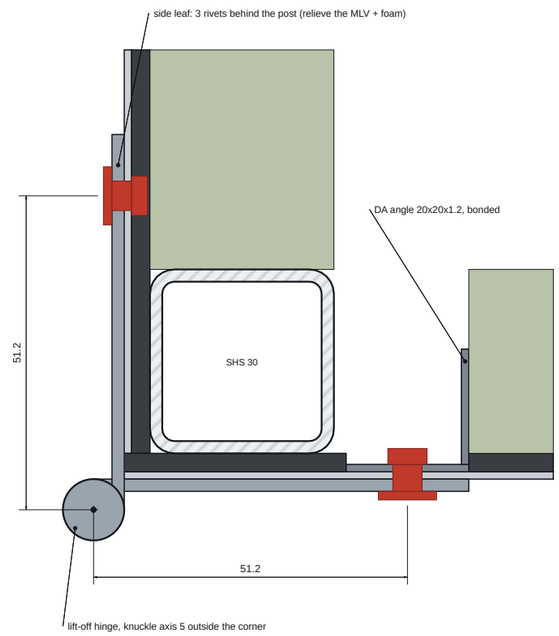
HARDWARE - hole patterns assume these parts (confirm before cutting)

ITEM	QTY	HOLES / NOTE
Lift-off butt hinge, stainless	5	100 long; rivet line 51.2 mm from the knuckle axis on both leaves (drill the leaves to suit), 3 x D5 per leaf at 35 pitch
Adjustable toggle latch, stainless	4	base 4 x D5 on 28 x 18, first row 52.2 mm behind the door face (detail L); hook engages the KB1 slot
Latch keeper KB1 (made, sheet 4)	4	riveted to door + stiffener, 2 x D5
Pull handle 160 c/c	2	2 x M5 from inside, through door + DA stiffener
Blind rivet 4.8 mm, stainless	54	4.8 x 8 (grip 3-5) for hinges and KB1, 4.8 x 6 for the latch bases; grips: door leaf 4.4, side leaf 3.2, KB1 3.6
M6 x 20 flanged button head	140	skins, strips and skirts into the frame rivnuts
Polycarbonate 6 mm, 180x240	2	drilled 8 x D5 to the WR1 pattern
EPDM 10 mm spacer frame + 8 x M4 x 35	1	window clamp: skin / pane / spacer / pane / WR1
Exhaust flanged spigot D150	1	4 x M5 on PCD 190 at 45 deg (RP1)
Brush cable grommet 200 x 40	1	4 x M4 on 224 x 64 (RP2)
16 mm drain bulkhead barb	1	D20.5 in SK2
Office air ports (printed P9)	2	D64 + 4 x D3.4 on the axes at r 38 per port (SR3); M3 x 10 button heads into the adaptor

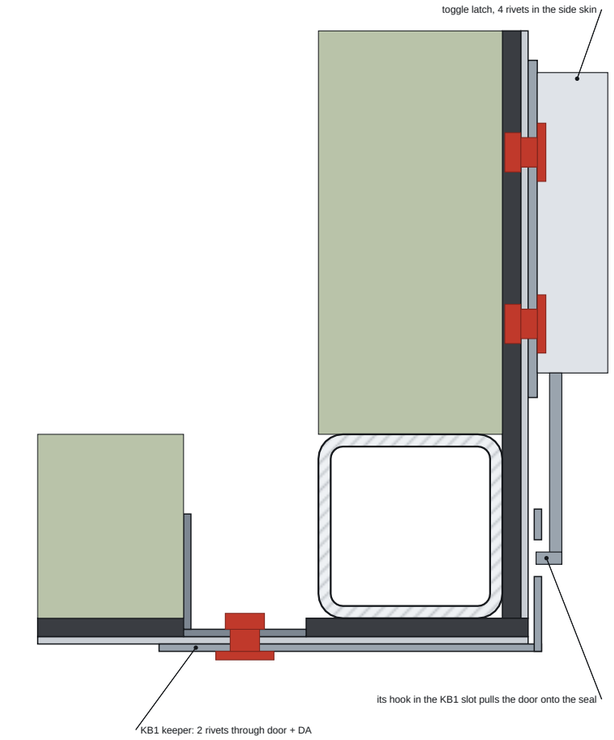
ASSEMBLY

1 Coat all parts. Bond MLV inside the skins and doors: punch it D8 at the frame screws (it is the gasket), clear it 3 mm round rivets and M4/M5 holes.
2 Doors: bond the DA stiffener frame inside (legs inboard, 2 mm inside the frame opening), rivet the hinges and KB1 keepers through skin + angle, fit handles.
3 Lower door: glazing unit - outer pane, EPDM spacer, inner pane, WR1 - clamped with 8 x M4 from outside.
4 Side skins bottom to top, then the JS strips over the joints (acoustic sealant under each strip edge).
5 Rear skins + JR strip, exhaust spigot and cable grommet; top skin last (it overlaps the side and rear skins).
6 Skirts SK1-SK3 up into the base-rail rivnuts. Hang the doors on the lift-off hinges, rivet the latches to the right-hand skins and adjust them to pull the doors 3 mm onto the seal.

H HINGE CORNER - PLAN 1:1

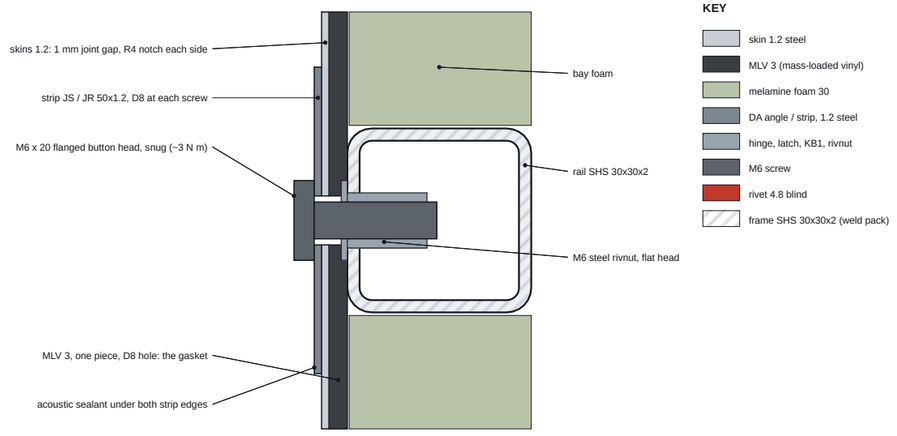


L LATCH CORNER - PLAN 1:1



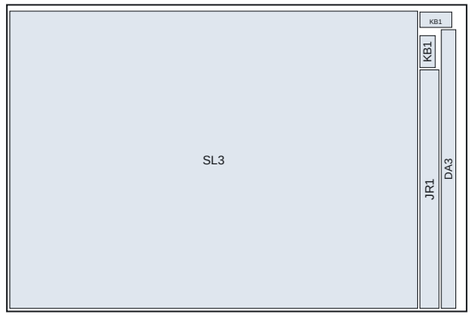
Latch rivets 52.2 and 80.2 behind the door face; KB1 rivets 46.2 in from the side.

J SKIN JOINT ON A RAIL - SECTION THROUGH A STRIP SCREW 1:1

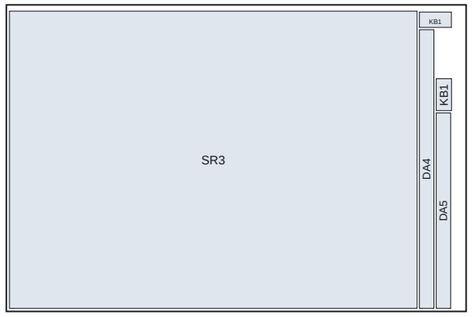


PROJECT	SRA-16 Silent AC Rack - laser-cut sheet-metal parts
TITLE	19in rail, hardware, assembly, details
DOC NO.	CP-SRA16-SMP-001 SHEET 5 OF 6 REV C
SCALE	as noted @ A3 UNITS mm LAYOUT v1.3.1
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	FOR CUTTING - DXF is the master; confirm hardware (sheet 5)

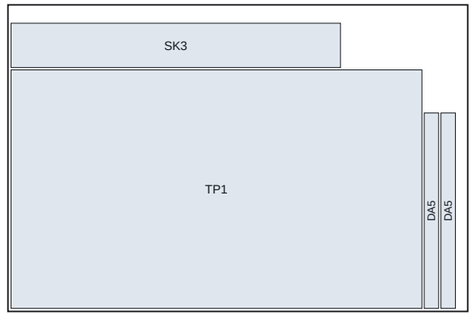
NESTING ON 1200 x 800 BLANKS (MaxRects, 6 mm gap, 8 mm margin) - cad/dxf/nest/



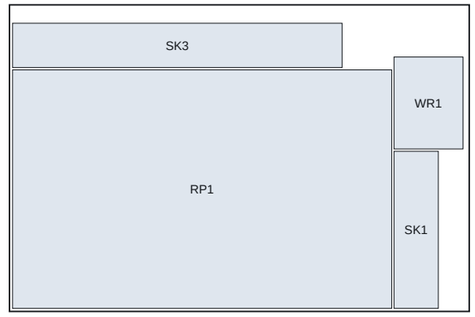
steel 1.2mm sheet 1 - 93% used



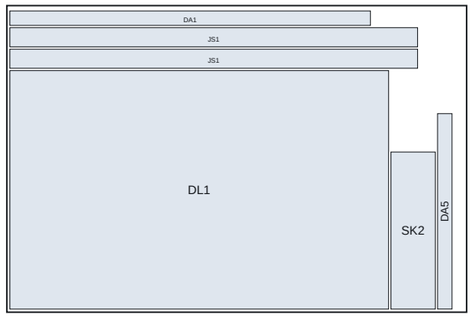
steel 1.2mm sheet 2 - 92% used



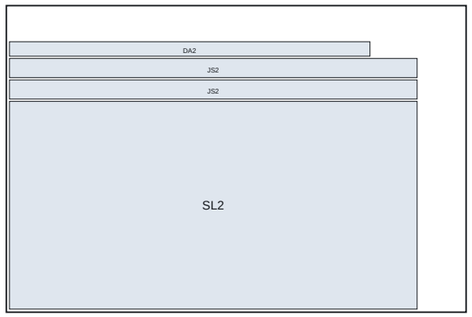
steel 1.2mm sheet 3 - 84% used



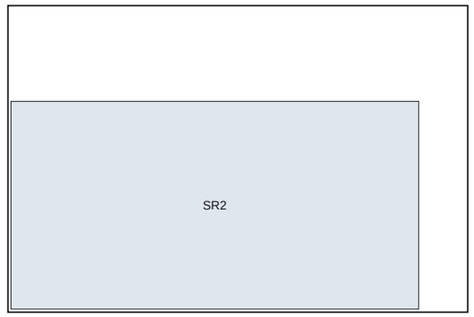
steel 1.2mm sheet 4 - 84% used



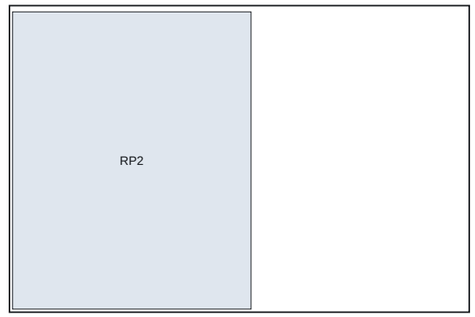
steel 1.2mm sheet 5 - 86% used



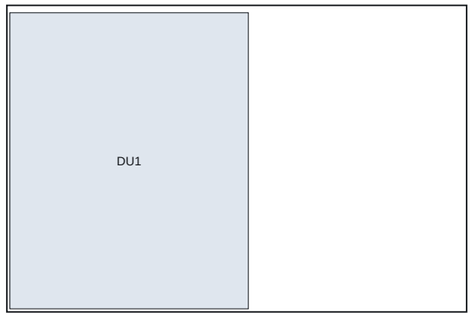
steel 1.2mm sheet 6 - 75% used



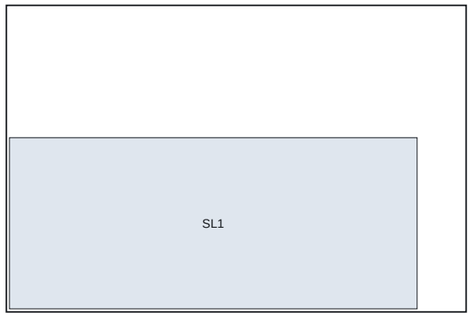
steel 1.2mm sheet 7 - 60% used



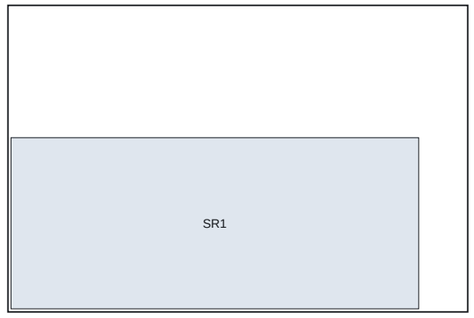
steel 1.2mm sheet 8 - 50% used



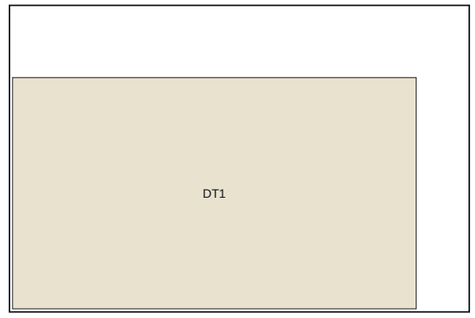
steel 1.2mm sheet 9 - 50% used



steel 1.2mm sheet 10 - 49% used



steel 1.2mm sheet 11 - 49% used



SS304 1.2mm sheet 1 - 66% used



plate 8mm sheet 1 - 4% used



plate 3mm sheet 1 - 0% used



steel 2mm sheet 1 - 14% used

NOTES

Blanks are the 1200 x 800 sheet / bed size. Each nest DXF has the parts on CUT and the blank on SHEET.

11 steel 1.2 blanks: the big skins cannot share a blank (no two fit together), so their offcuts carry the strips, stiffeners, keepers and skirts.

PD1 (8 mm), PC1 (3 mm) and RR1 (2 mm) are small: cut them from offcuts or order them as cut parts; their nests are shown for completeness.

Parts may turn 90 deg (no grain on coated steel). 6 mm between parts, 8 mm from the blank edge.

PROJECT	SRA-16 Silent AC Rack - laser-cut sheet-metal parts
TITLE	Nesting on 1200 x 800 blanks
DOC NO.	CP-SRA16-SMP-001 SHEET 6 OF 6 REV C
SCALE	1:16 @ A3 UNITS mm LAYOUT v1.3.1
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	FOR CUTTING - DXF is the master; confirm hardware (sheet 5)